ThreatMatrix AI — Progress Report #1

> **Project Title:** AI-Powered Network Anomaly Detection and Cyber Threat Intelligence System utilizing Machine Learning and Real-Time Traffic Analysis
> **Report Date:** February 19, 2026
> **Reporting Period:** February 10 – February 19, 2026
> **Submitted By:** Kidus Abdula (Lead Software Engineer & Systems Architect)
> **Status:** ◍ On Track — Active Development

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1. Executive Summary

Development of ThreatMatrix AI has progressed on an aggressive two-month sprint cycle. The first two weeks have been dedicated to **architectural finalization, comprehensive documentation, and foundational implementation**. The system architecture has been fully specified across a 5-part master documentation suite totaling ~210KB of technical specification. Simultaneously, the backend infrastructure, database schema, packet capture engine, and ML training pipeline have entered active development, achieving approximately **50% of core implementation** milestones. The project remains firmly on schedule for full delivery by April 2026.

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2. Overall Progress Dashboard

Dimension	Progress	Status
Documentation & Specification	**80%**	☑ Ahead of Schedule
System Implementation	**50%**	☑ On Track
ML Model Development	**40%**	☑ On Track
Frontend / UI Development	**30%**	☑ On Track
Testing & Validation	**20%**	☑ On Track
Deployment & DevOps	**30%**	☑ On Track

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3. Documentation Status (80% Complete)

Documentation has been treated as a **first-class deliverable**, produced concurrently with architectural decision-making. The following documents are complete or in advanced draft:

3.1 Completed Documents

#	Document	Pages	Status	Description

#	Document	Pages	Status	Description
1	**Master Doc Part 1: Executive Strategy & Business Case**	12	☑ Complete	Problem statement, market analysis, competitive landscape, SWOT analysis, monetization model, target customer profiles, advisor impression strategy, risk register
2	**Master Doc Part 2: System Architecture & Infrastructure**	15	☑ Complete	Three-tier architecture blueprint, VPS deployment topology, full PostgreSQL schema (8 tables with indexes), REST API specification (40+ endpoints), WebSocket event contracts, Redis Pub/Sub design, RBAC matrix, technology stack rationale
3	**Master Doc Part 3: Module Specifications & UI/UX Design System**	14	☑ Complete	CSS design system (color palette, typography, animation keyframes), all 10 module specifications with component inventories, War Room layout blueprint, navigation architecture, priority classification
4	**Master Doc Part 4: ML Pipeline, LLM Integration & Data Strategy**	18	☑ Complete	Three-model ML strategy (Isolation Forest, Random Forest, Autoencoder) with training configurations, 40+ feature engineering specifications, ensemble scoring formula, LLM gateway architecture with multi-provider routing, prompt templates, budget projections, threat intelligence feed integration, end-to-end data flow
5	**Master Doc Part 5: Development Timeline & Deployment Guide**	16	☑ Complete	Week-by-week sprint plan, full monorepo directory structure, task assignment matrix, Docker Compose production configuration, testing strategy, demo day script, version milestone roadmap

Total documentation volume: ~210 KB across 5 interconnected documents (~75 pages equivalent)

3.2 Documents In Progress

#	Document	Progress	Target Date
6	API Reference (OpenAPI/Swagger)	40% — auto-generated from FastAPI	Week 4
7	ML Evaluation Report	20% — pending model training completion	Week 5
8	User Manual	10% — outline drafted	Week 7
9	Business Plan & Revenue Model	30% — business manager drafting	Week 6

#	Document	Progress	Target Date
10	Final Presentation Slides	15% — structure defined	Week 7

3.3 Documentation Architecture

The master documentation follows **Domain-Driven Documentation (DDD)** — each document owns a bounded context and cross-references related documents via navigable links. This ensures the documentation scales alongside the codebase without duplication or drift.

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4. Implementation Status (50% Complete)

4.1 Backend Infrastructure

Component	Status	Details
Project Monorepo Structure	☑ Complete	`/backend` (Python/FastAPI), `/frontend` (Next.js 16), `/docs`, Docker configs
PostgreSQL Schema	☑ Complete	8 tables designed and specified: `users`, `network_flows`, `alerts`, `threat_intel_iocs`, `ml_models`, `capture_sessions`, `pcap_uploads`, `llm_conversations`, `system_config`, `audit_log`
Database Migrations	☑ Complete	Alembic migration pipeline configured, initial migration generated
FastAPI Application Shell	☑ Complete	App factory, CORS, middleware, OpenAPI auto-docs, health endpoint
Authentication Service	☑ Complete	JWT (access + refresh tokens), bcrypt password hashing, login/register/refresh endpoints
RBAC Authorization	☑ Complete	4 roles (admin, soc_manager, analyst, viewer), permission decorators
Docker Compose Stack	☑ Complete	PostgreSQL 16, Redis 7, FastAPI backend, capture engine, ML worker, Nginx — all containerized
Redis Integration	☑ Complete	Pub/Sub channels configured, connection pooling, caching middleware
Flow Data API	🔨 In Progress	CRUD endpoints for network flows, aggregation queries, search
Alert Management API	🔨 In Progress	Alert lifecycle endpoints, severity filtering, assignment

Component	Status	Details
WebSocket Server	🔨 In Progress	Event broadcasting architecture, client connection management

4.2 Capture Engine

Component	Status	Details
Scapy Packet Sniffer	☑ Complete	Raw packet capture on configurable network interfaces with BPF filter support
Flow Aggregator	☑ Complete	5-tuple flow assembly (src_ip, dst_ip, src_port, dst_port, protocol) with configurable timeout (30s/120s)
Feature Extraction Pipeline	🔨 In Progress	40+ features per flow — basic features (duration, bytes, packets) complete; statistical and behavioral features in development
Redis Publisher	☑ Complete	Completed flows published to `flows:live` channel
PCAP Ingestion	📋 Planned	Week 5 deliverable

4.3 Machine Learning Pipeline

Component	Status	Details
NSL-KDD Dataset Loader	☑ Complete	Download, parse, clean, encode, normalize, stratified split (80/20)
Feature Preprocessing	☑ Complete	Label encoding, one-hot encoding, StandardScaler normalization
Isolation Forest	🔨 In Progress	Model architecture defined, hyperparameter grid established, initial training runs executing
Random Forest Classifier	🔨 In Progress	Multi-class configuration (Normal/DoS/Probe/R2L/U2R), class balancing weights set
Autoencoder (TensorFlow)	📋 Planned	Architecture specified (64→32→16→32→64), training scheduled for Week 3
Ensemble Scorer	📋 Planned	Weighted composite scoring formula defined, implementation Week 3-4
Model Evaluation Framework	🔨 In Progress	Metrics pipeline (accuracy, precision, recall, F1, AUC-ROC, confusion matrix) partially implemented
CICIDS2017 Dataset	📋 Planned	Secondary validation dataset, Week 5

4.4 LLM Integration

Component	Status	Details
LLM Gateway Service Architecture	☑ Complete	Multi-provider router designed: DeepSeek V3 → GLM-4-Flash → Groq Llama 3.3
Provider API Integration	🔨 In Progress	DeepSeek V3 client implemented and tested; GLM and Groq in progress
Prompt Template Library	☑ Complete	7 templates: alert analysis, daily briefing, IP investigation, Amharic translation, chat, playbook, executive summary
Budget Tracking System	📋 Planned	Token counting and cost logging, Week 4
Response Caching	📋 Planned	Redis-based caching with TTL, Week 4

4.5 Frontend / Command Center

Component	Status	Details
Next.js 16 Project Init	☑ Complete	App Router, TypeScript strict mode, project configuration
Design System (CSS)	☑ Complete	Full CSS variable system — color palette (cyber-cyan, threat-red, deep-black), typography (JetBrains Mono + Inter), glassmorphism mixins, animation keyframes
Layout Shell	☑ Complete	Sidebar navigation (icon-only, 10 modules), TopBar (threat badge, notifications, user), StatusBar (system health)
GlassPanel Component	☑ Complete	Reusable glassmorphism container with backdrop blur
War Room — Page Structure	🔨 In Progress	Grid layout defined, component slots allocated
War Room — ThreatMap	🔨 In Progress	Deck.gl + Maplibre integration, dark basemap rendering
War Room — MetricCards	🔨 In Progress	Animated counter cards with sparkline integration
Remaining 9 Module Pages	📋 Planned	Weeks 4-7

4.6 Threat Intelligence

Component	Status	Details
AlienVault OTX Integration	🔨 In Progress	API client implemented, pulse/IOC retrieval working
AbuseIPDB Integration	📋 Planned	IP reputation lookup, Week 4
VirusTotal Integration	📋 Planned	Domain/hash analysis, Week 4
IOC Correlation Engine	📋 Planned	Cross-reference IOCs with live flows, Week 4-5

5. Technical Decisions Finalized

The following architectural decisions have been formally evaluated and locked:

Decision	Choice	Rationale
Frontend Framework	**Next.js 16**	Team expertise from enterprise projects (Pana ERP v3.0), SSR/SSG capabilities, API routes
Backend Framework	**FastAPI (Python)**	Async support, auto-generated OpenAPI docs, native ML ecosystem compatibility
Database	**PostgreSQL 16**	Enterprise-grade, JSONB support for feature vectors, INET type for IP addresses, full-text search
Cache / Pub-Sub	**Redis 7**	Real-time alert broadcasting, LLM response caching, rate limiting, session management
Map Visualization	**Deck.gl + Maplibre GL**	Zero-cost (fully open source), WebGL-powered rendering, no API key dependency
Real-Time Strategy	**WebSocket + Redis Pub/Sub**	Multi-client broadcasting, capture-backend decoupling, message replay capability
ML Models	**Isolation Forest + Random Forest + Autoencoder**	Unsupervised + Supervised + Deep Learning — demonstrates three distinct ML paradigms
LLM Providers	**DeepSeek V3 + GLM-4-Flash + Groq**	Cost-optimized: complex analysis ($0.14/M) + bulk tasks ($0.01/M) + real-time ($0.06/M)
Capture Strategy	**VPS host capture (primary) + local agent (optional)**	Legally safe (own infrastructure), always-on, real production traffic
Deployment	**Docker Compose on VPS + Vercel (frontend)**	Single-command deployment, SSL via Let's Encrypt, global CDN for frontend
i18n	**next-intl**	Battle-tested Next.js internationalization for Amharic/English

6. Risk Status

Risk	Probability	Impact	Status	Mitigation Active

Risk	Probability	Impact	Status	Mitigation Active
Timeline overrun	Medium	Critical	🟢 Controlled	Aggressive agile sprints, each version is demoable
ML accuracy below threshold	Medium	High	🟢 Controlled	Three models provide redundancy; NSL-KDD is well-studied
LLM budget exceeded	Low	Medium	🟢 Controlled	Budget tracking designed, projected cost ~$0.50/month
VPS downtime during demo	Low	Critical	🟡 Monitoring	Docker restart policies, backup demo video planned
Scope creep	Medium	High	🟢 Controlled	Master documentation acts as scope boundary

7. Budget Status

Item	Allocated	Spent to Date	Remaining
LLM APIs (DeepSeek + GLM + Groq)	$120	$3.40	$116.60
VPS Infrastructure	$0 (owned)	$0	—
Domain / Hosting	$0 (Vercel free)	$0	—
Reserve	$50	$0	$50
Total	**$170**	**$3.40**	**$166.60**

Budget utilization is significantly under projection — LLM costs are lower than estimated due to efficient prompt engineering and response caching during development.

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8. Next Sprint Objectives (Week 3-4: Feb 24 — Mar 9)

Priority	Task	Owner	Deliverable
🟢 P0	Complete ML model training (all 3 models on NSL-KDD)	Lead Architect	Trained models with evaluation metrics
🟢 P0	Real-time inference pipeline (capture → ML → scoring → alerts)	Lead Architect	End-to-end anomaly detection working
🟢 P0	LLM Gateway: all 3 providers integrated with fallback routing	Lead Architect	AI Analyst backend functional
🟢 P0	War Room: ThreatMap, LiveAlertFeed, TrafficTimeline, MetricCards	Full-Stack Dev	War Room visually complete with live data
🟡 P1	Alert Console: table, detail drawer, severity filtering	Full-Stack Dev	Alert management page working

Priority	Task	Owner	Deliverable
P1	Threat intel feed sync (OTX + AbuseIPDB)	Lead Architect	IOC database populated
P1	AI Analyst chat UI with streaming responses	Full-Stack Dev	Conversational AI interface working
P2	Business plan first draft	Business Manager	Revenue model document
P2	Test scenario definitions + attack simulation scripts	Tester	Test plan with nmap/hping3 scenarios

Critical Milestone (End of Week 4): v0.4.0 — The system must demonstrate a complete intelligence loop: live traffic capture → ML anomaly detection → automated alerts → LLM-powered threat narratives → real-time dashboard display. This constitutes a **fully functional MVP**.

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9. Team Status

Member	Role	Current Focus	Blockers
Kidus Abdula	Lead Architect	ML pipeline training, capture engine feature extraction	None
Team Member 2	Full-Stack Dev	War Room UI components, design system polish	Waiting for WebSocket endpoint (resolved this sprint)
Team Member 3	Business Manager	Market research, legal framework for traffic capture	Awaiting legal consultation feedback
Team Member 4	Tester	Test plan documentation, dev environment setup	None

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10. Appendix: Key Artifacts Produced

Artifact	Location	Type
Master Documentation (5 parts)	`/docs/MASTER_DOC_PART[1-5]_*.md`	Technical Specification
Database Schema (SQL)	Defined in Part 2, Section 4	Schema Definition
API Specification (40+ endpoints)	Defined in Part 2, Section 5	API Contract
ML Model Configurations	Defined in Part 4, Sections 4-6	ML Specification
LLM Prompt Templates (7)	Defined in Part 4, Section 9.2	Prompt Engineering
Docker Compose Config	Defined in Part 5, Section 6.3	DevOps
UI Design System (CSS)	Defined in Part 3, Section 1	Design Specification
Module Specifications (10)	Defined in Part 3, Sections 2-11	Functional Specification

Artifact	Location	Type
Project Directory Structure	Defined in Part 5, Section 2	Architecture
Weekly Sprint Plan (8 weeks)	Defined in Part 5, Section 3	Project Management

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This progress report reflects the project status as of February 19, 2026.
Next progress report scheduled: March 3, 2026 (End of Week 4 — MVP milestone)

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